1. **DATA OVERVIEW**
   - Data sources and download process:
     - Cardiovascular disease mortality, smoking prevalence, and high blood pressure data were obtained from the U.S. Chronic Disease Indicators (CDI, 2023 release) and downloaded as a CSV from the CDC website.
     - Ozone exposure data were obtained from the Daily Census Tract-Level Ozone Concentrations dataset (2011–2014), and downloaded as a CSV from the CDC website.
   - Data generation:
     - The CVD data represent a census of reported age-adjusted mortality rates at the state level.
     - Ozone data aggregated from census tracts by *year* and *statefips* using the average of *ds_o3_pred.*
   - Systematically excluded groups:
     - Non-reporting or incomplete states/years: States or years with missing or incomplete CVD or ozone data were excluded from the analysis.
     - Subpopulations within states: The data are aggregated at the state level and do not include finer geographic units (e.g., counties) or individual-level demographics, so groups with localized high or low exposure may be masked.
     - Non-U.S. regions: Data from U.S. territories or non-state regions were not included.
   - Data granularity:
     - CVD data: Each row represents a state in a specific year (state-year) with columns for the CVD question, age-adjusted mortality rate, and year. This smooths over within-state variation and prevents assessment of individual-level risk factors or localized hotspots of CVD mortality.
     - Ozone exposure data: Each row corresponds to a state in a given year with average ozone concentration. This captures general state-level exposure patterns but cannot detect high-exposure events or local variation that might influence health outcomes.
   - Potential data concerns:
     - Selection bias: States with missing data were excluded.
     - Measurement error:
       - Modeled ozone averages may not perfectly represent true exposure.
       - Aggregated CVD rates may smooth over local variation.
     - Unavailable features: Individual demographics (e.g., age, sex), socioeconomic factors, other pollutants are not included, limiting causal inference and subgroup analyses.
     - Missing data: Rows with missing CVD or ozone values were dropped, which could introduce bias if missingness is not random.

- Data cleaning and preprocessing:
  - For research question 1, we cleaned the data by renaming columns for clarity, filtering the CVD dataset to include only relevant state-year observations from 2011–2014, and dropping rows with missing values. CVD rates and ozone concentrations were converted to numeric types, and the datasets were merged on state and year. We aggregated values to the state-year level to remove duplicates and computed year-specific ozone quartiles for comparison. These steps create a clean, consistent dataset suitable for statistical testing, but aggregation and missing-data removal may smooth over short-term peaks and exclude certain states, limiting sensitivity to extreme exposures and slightly affecting generalizability.
  - For research question 2, we cleaned the data by restricting the dataset to rows that are at the state or state-equivalent level (jurisdictions with two-letter abbreviations, excluding national rows), have StratificationCategory1 / Stratification1 = "Overall", and come from the years 2011, 2013, 2015, 2017, 2019, and 2021.Additionally, we required non-missing values for both smoking and blood pressure. Then, we aggregated to one row per state–year and dropped any remaining missing values. This outputted a panel with 310 state–year observations across 52 jurisdictions and six years. Each state contributed repeated measurements over time, making the state the natural grouping variable for a hierarchical model.

2. RESEARCH QUESTIONS

- Research question 1: Are cardiovascular disease (CVD) rates significantly higher in regions with elevated ozone exposure compared to regions with lower ozone levels?
  - Answering this question can inform public health policy and regulatory decisions, such as targeting air quality interventions, issuing health advisories in high-ozone regions, or prioritizing resources for CVD prevention in areas with elevated pollution.
  - We use multiple hypothesis testing / decision making to compare CVD rates across ozone exposure quartiles, combined with multiple hypothesis testing corrections (Bonferroni and Benjamini-Hochberg). This method is appropriate because our data are aggregated, continuous, and independent across state-year observations, and we are primarily interested in detecting differences in mean CVD rates across defined exposure groups. The quartile approach allows us to examine patterns across the exposure distribution rather than assuming a single cutoff.
  - Limitations: It does not control for potential confounders (e.g., other pollutants, demographics, or socioeconomic factors), so observed differences may reflect correlations rather than causation. It also assumes independence and roughly

normal distributions, which may not fully hold for state-year aggregates. The method may underperform with small sample sizes or small effect sizes, and is less sensitive to nonlinear or temporal relationships between ozone and CVD. Under these circumstances, it could fail to detect meaningful associations or overstate the significance of spurious differences.

- Research question 2: How is adult cigarette smoking prevalence associated with high blood pressure prevalence across U.S. states over time, and how much of the variation in high blood pressure can be explained by persistent differences between states?
    - Answering this question allows for many real-world health care decisions to be made within the US. For example, in states where smoking and high blood pressure are more prevalent, more funding for tobacco control and hypertension management could be allocated. Integrated care programs could also be established in states where there appears to be higher correlation between smoking and high blood pressure.
    - We decided to use a Bayesian hierarchical model for this question. This method is appropriate for our question because of the multi-level aspect of our data. Each state is not independent of another given the fact that healthcare systems and policies can affect one another. We want each state to have its own baseline while also maintaining and learning the overall trends across all states. Additionally, we want to know the association between higher blood pressure and smoking rather than if there is a causal effect between the two. The hierarchical model provides us with credible intervals for smoking and time trends, as well as the variance between and within states.
    - Limitations: As with the first question, our model does not account for all possible confounders. These factors can influence smoking and/or blood pressure. This means that our model might be overstating the association between the two. We are also looking at state averages rather than individual people, so it is important to remember that even if a state with a higher smoking rate has a higher BP prevalence, it doesn't mean that an individual smoker from the state will have increased BP risk compared to a nonsmoker from the same state. In our model, we are also assuming a constant smoking slope, but in reality this association could be non-linear.

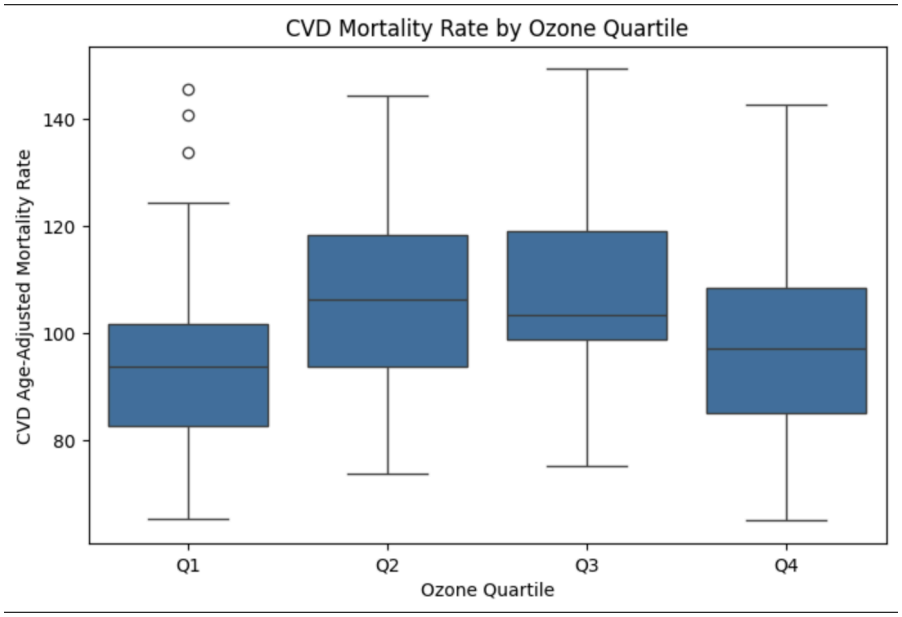
3. PRIOR WORK

- A landmark study by Samet et al. (2000, NMMAPS) addressed how short-term air pollution, including ozone, affects daily mortality across multiple U.S. cities.
    - Relevance: This is directly relevant to our research question, which investigates whether regions with higher ozone exposure have elevated CVD mortality rates. Both studies focus on population-level associations between ozone and health

outcomes, although Samet et al. examined daily short-term effects, while our analysis considers state-year aggregated long-term average exposure.
    - Methodological comparison: Samet et al. used time-series models with fixed effects and weather adjustments, which allow precise control for temporal confounding, autocorrelation, and short-term fluctuations. In contrast, our approach uses aggregated quartile comparisons with Welch t-tests and multiple hypothesis corrections, which are simpler and more suitable for a Data 102-level analysis. Our method does not adjust for daily weather variation or temporal autocorrelation, so our findings capture broader, average associations rather than precise short-term effects. The simpler methods may underestimate or overestimate effect sizes compared to Samet et al., and residual confounding is more likely, but they still provide meaningful insights into general patterns of ozone-CVD relationships.
- An epidemiological review by Gao et al. (2023) examined the relationship between tobacco use and hypertension across populations.
    - Relevance: This work is directly relevant to our research, which studies the relationship between adult cigarette smoking prevalence and high blood pressure prevalence across the U.S. states over time. Both analyses focus on population-level associations between smoking and blood pressure outcomes, but Gao et al. more summarizes individual-level and cohort-based findings.
    - Methodological comparison: Gao et al. used cohort designs, regression-based adjustments, and stratified analyses to estimate the associations between smoking status and hypertension risk. The study also controls for individual-level confounders such as age, sex, and comorbidities. Our research uses a Bayesian hierarchical model with state-level varying intercepts and global slopes for smoking and time. It allows us to quantify differences between states and overall trends, rather than individual risk, which highlights how smoking and blood pressure can co-vary across regions and over time at the population level.

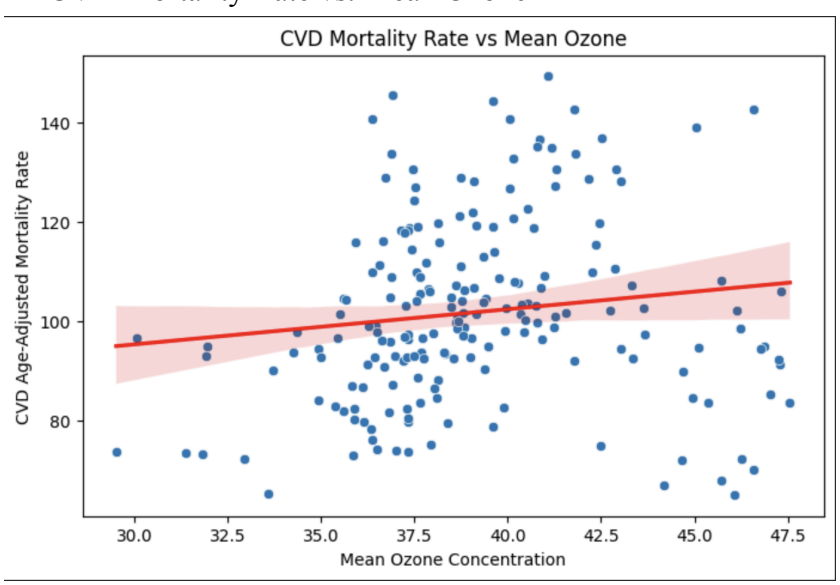
4. EDA

- CVD Mortality Rate by Ozone Quartile



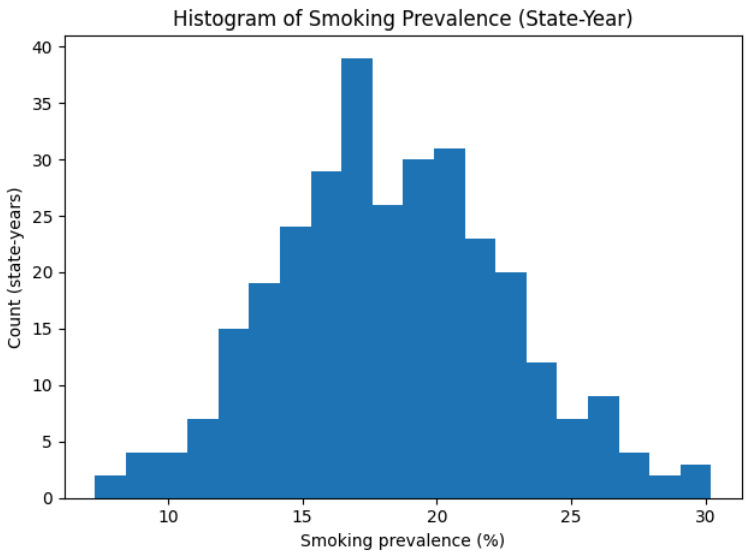
Median CVD mortality is lowest in Q1 and higher in Q2 and Q3, while Q4 does not consistently show the highest mortality. Substantial overlap across quartiles indicates high within-group variability. This plot directly supports the hypothesis-testing design by showing differences across ozone exposure groups. It explains why comparisons involving Q2 and Q3 are more likely to be significant, while Q4 comparisons often are not, reinforcing the interpretation of the statistical results.

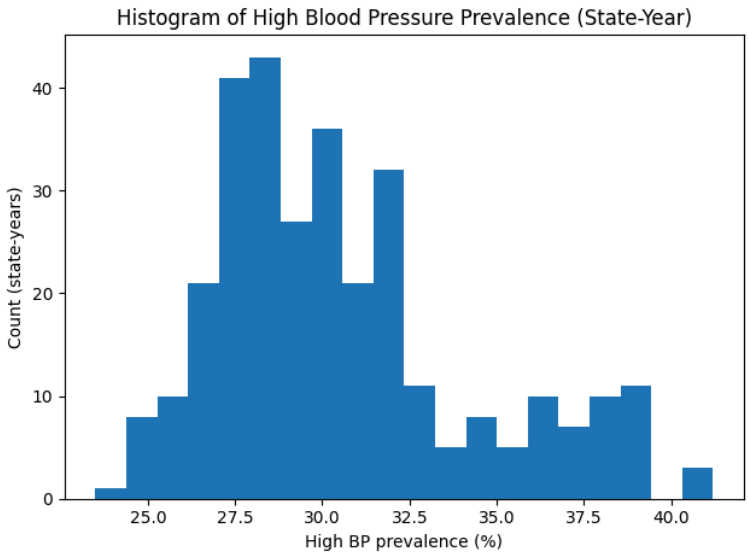
- CVD Mortality Rate vs. Mean Ozone



There is a weak positive relationship between mean ozone concentration and CVD mortality, with higher ozone levels associated with slightly higher mortality rates. The large spread around the trend line indicates substantial variability and a weak linear association. This plot suggests a possible association between ozone exposure and CVD mortality, while also highlights why formal hypothesis testing and multiple-testing correction are necessary due to noise and confounding.

- Histograms of Smoking and High Blood Pressure Prevalence

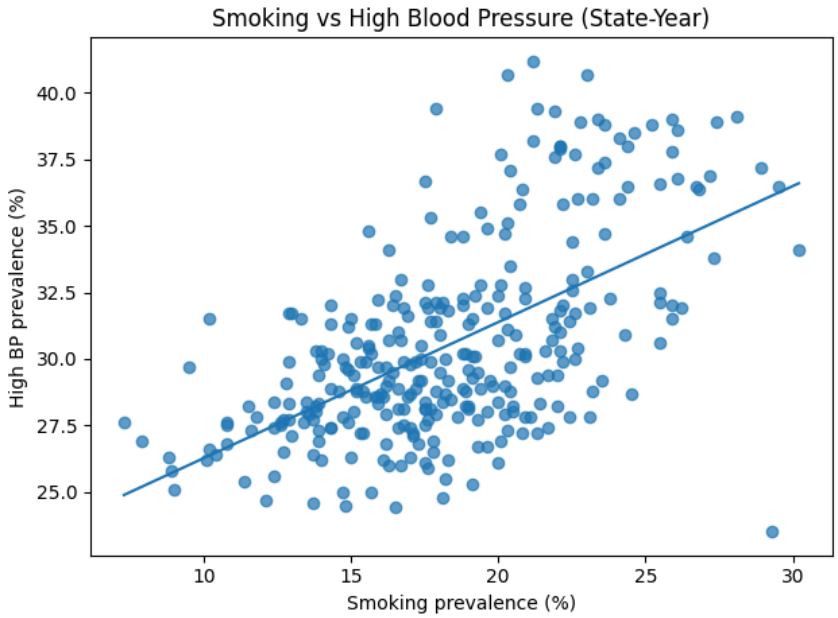




These two histograms show us that both of these variables are roughly centered around a certain value, giving them an almost unimodal shape. This is important in allowing us to model them as approximately normal at the state-year level. It also tells us that we have

reasonable spread in the exposure and outcome side, making it possible to detect patterns. From here, we know that we can use our hierarchical model to formalize the potential relationship given by our research question.

- High Blood Pressure Prevalence vs. Smoking Prevalence



This scatterplot illustrates a moderate positive relationship between high blood pressure prevalence and smoking prevalence. There is still a considerable amount of variation, seen by the wide spread of points around the trend line. However, this moderate relationship still indicates potential correlation between the two variables, motivating us to continue looking deeper into the relationship between the two. By using a hierarchical model, we can formalize the trends that we potentially see here and also account for differences among states.

5. RESEARCH QUESTION 1 AND MULTIPLE HYPOTHESIS TESTING / DECISION MAKING

- **Methods**
  - We test six pairwise hypotheses comparing mean CVD rates across ozone exposure quartiles (Q1 - Q4), where Q1 represents the lowest ozone exposure and Q4 the highest. For each comparison, the null hypothesis is that mean CVD mortality rates are equal across the two ozone exposure groups:
    - H_{01}: Mean CVD rate in Q4 equals mean CVD rate in Q1
    - H_{02}: Mean CVD rate in Q3 equals mean CVD rate in Q1
    - H_{03}: Mean CVD rate in Q2 equals mean CVD rate in Q1
    - H_{04}: Mean CVD rate in Q4 equals mean CVD rate in Q2

    - H_{05}: Mean CVD rate in Q4 equals mean CVD rate in Q3
    - H_{06}: Mean CVD rate in Q3 equals mean CVD rate in Q2

    Testing multiple hypotheses instead of a single comparison is necessary because ozone exposure is continuous and lacks a natural binary cutoff; focusing on only one contrast (such as highest vs lowest exposure) could miss important patterns such as nonlinear or threshold effects. By examining multiple pairwise comparisons across the exposure distribution, we can assess whether CVD mortality differs at moderate versus high ozone levels and obtain a more comprehensive picture of the ozone - CVD relationship, while accounting for the increased risk of false positives through multiple hypothesis testing corrections.
  - For the key comparison between the highest and lowest ozone exposure groups (Q4 vs Q1), we test the null hypothesis that the mean age-adjusted CVD mortality rates are equal against the specific two-sided alternative hypothesis
    H_{11}: $\mu_{Q4} - \mu_{Q1} = 5$ deaths per 100,000 population.
    The value of 5 was chosen because it closely matches the observed difference in the data and represents a small but epidemiologically meaningful change in population-level mortality.
    Using a Welch two-sample t-test with sample sizes $n_{Q4} = 48$ and $n_{Q1} = 52$, a pooled standard deviation of approximately 18.7, and significance level $\alpha = 0.05$, the implied effect size is Cohen's $d \approx 0.268$.
    A power analysis based on this alternative yields an estimated power of about 0.263, indicating that the test has a low probability of detecting an effect of this magnitude, which helps explain why some high-ozone versus low-ozone comparisons fail to reject the null despite non-zero mean differences.
  - This test is appropriate because:
    - The outcome (age-adjusted CVD mortality rate) is continuous,
    - Comparisons are between independent groups,
    - Variances and sample sizes differ across quartiles.

    The Welch t-test performs well with moderate sample sizes, which fits the state-year aggregated data used here. Pairwise mean comparisons directly address whether CVD rates differ across specific ozone exposure levels, making this approach well aligned with the research question.
  - To account for testing multiple hypotheses, we use two different correction methods that control different types of error rates.
    - The Bonferroni correction controls the family-wise error rate (FWER) by multiplying each p-value by the total number of tests. This approach is very conservative and prioritizes avoiding any false discoveries.
    - The Benjamini-Hochberg (BH) procedure controls the false discovery rate (FDR) by adjusting p-values based on their rank.

Using both methods allows us to compare a strict error-control approach (FWER) with a more powerful, exploratory approach (FDR), which is especially appropriate when assessing multiple related comparisons rather than making a single definitive claim.

  - Applying the Bonferroni correction, which strictly controls the probability of making even a single false positive, results in two statistically significant discoveries (Q3 vs Q1 and Q2 vs Q1). The BH procedure, which controls the expected proportion of false positives among rejected hypotheses, also yields two discoveries in this analysis. Although both methods lead to the same number of significant results here, they differ in interpretation: Bonferroni provides stronger protection against false positives at the cost of reduced power, while BH allows more flexibility and higher sensitivity. For this research question, which aimed at identifying patterns in population-level associations between ozone exposure and CVD mortality rather than making a single definitive causal claim, FDR control is more appropriate, as it balances error control with the ability to detect meaningful associations across multiple related comparisons.

- **Results**
  - Six pairwise t-tests compared CVD rates across ozone quartiles. Moderate ozone exposure groups (Q2 and Q3) had significantly higher CVD rates than the lowest quartile (Q1), while the highest quartile (Q4) showed no consistent differences.
  - After multiple testing corrections, only Q2 vs Q1 and Q3 vs Q1 remained significant under FDR, and none were significant under the more conservative Bonferroni correction. Effect sizes were small to moderate (Cohen's d ~ 0.27 - 0.36), and low power (~26% for Q4 vs Q1) likely explains the non-significant results at high exposure.
  - Overall, results suggest a positive association between moderate ozone exposure and CVD mortality, but aggregation, missing data, and lack of individual-level covariates limit sensitivity and causal interpretation. Additional data on daily ozone peaks, demographics, and other pollutants could improve analysis.
- **Discussion**
  - After applying the multiple testing corrections, two discoveries remained significant under both the Bonferroni (FWER) and BH(FDR) procedures: Q3 vs Q1 and Q2 vs Q1. All other comparisons were no longer statistically significant. This is likely due to a combination of small effect sizes, limited sample sizes, and low statistical power, which reduce the ability to detect differences, particularly for the Q4 group despite its higher nominal ozone levels.
  - From the individual tests, we can say that moderate ozone exposure (Q2 and Q3) is associated with higher CVD mortality rates compared to the lowest exposure group (Q1). These findings suggest that public health interventions or monitoring

could focus on regions with moderate ozone levels, as they show a measurable increase in risk relative to low-exposure areas.

- From the results in aggregate, the pattern indicates that the relationship between ozone and CVD mortality is not strictly monotonic, since the highest ozone exposure group (Q4) does not consistently show higher CVD rates. This suggests that while moderate ozone levels are clearly associated with increased risk, other factors, such as regional demographics, healthcare access, or co-pollutants, may influence outcomes. Overall, the aggregate evidence supports targeted public health strategies rather than a simple assumption that higher ozone always corresponds to higher CVD risk.
- Limitations:
  - The data are ecological (state-year level), which prevents drawing individual-level causal conclusions and may mask within-state variation in ozone exposure and CVD risk.
  - Potential confounders such as NO_2, socioeconomic factors, and healthcare access were not fully controlled for, so observed associations may be partially influenced by these unmeasured variables.
  - There may be spatial and temporal dependence between observations, violating the independence assumption of the t-tests, which could slightly bias p-values.
  - Statistical power is limited, especially for comparisons involving the highest ozone quartile (Q4), making it harder to detect modest effects.
- To avoid p-hacking, all hypotheses and pairwise comparisons were pre-specified based on ozone quartiles before analysis, and quartiles were defined independently of the outcome. No subgroup selection or outcome-switching was performed after seeing the results, and we applied multiple hypothesis testing corrections (Bonferroni and Benjamini–Hochberg) to control for inflated FPRs. This ensures that the findings are robust and not the result of selective reporting or post-hoc choices.
- If more data were available,
  - We would conduct multi-pollutant regression to account for other air pollutants (e.g., PM2.5, NO_2) and isolate the independent effect of ozone on CVD mortality.
  - Longitudinal or fixed-effects models would allow us to control for unobserved, time-invariant differences between regions and better capture temporal trends.

- Our results are similar to Samet et al. (2000) in that both studies find a positive association between air pollution and CVD. In our analysis, moderate ozone exposure (Q2 and Q3) is linked to higher CVD rates compared to the lowest exposure group, which aligns with Samet et al.'s general finding that higher

pollutant levels increase mortality risk. However, our results differ in that the highest ozone quartile (Q4) did not consistently show elevated CVD rates after multiple testing corrections, whereas Samet et al. observed stronger effects at the highest pollution levels over short time periods. This discrepancy likely arises from several factors: our state-year aggregation smooths out short-term peaks in ozone, we did not explicitly control for weather or other pollutants, and our smaller sample sizes reduce statistical power to detect subtle effects. Overall, while the direction of the association is consistent, the magnitude and pattern of effects differ due to differences in data granularity, temporal scale, and modeling approach.

6. RESEARCH QUESTION 2 AND BAYESIAN HIERARCHICAL MODELING

- **Methods**

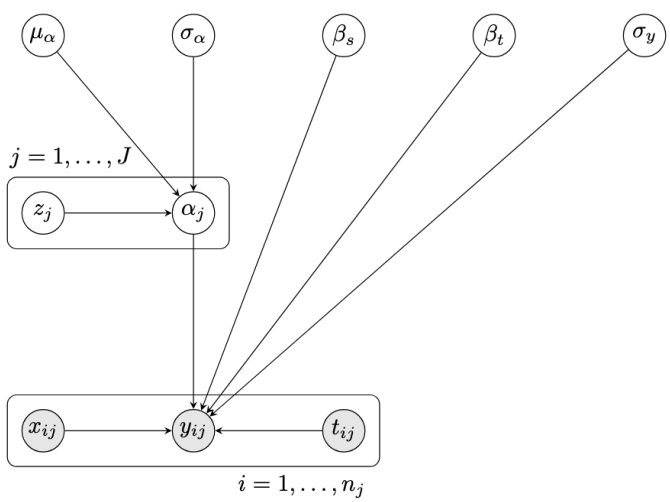


Figure 1: Hierarchical Bayesian model for the association between smoking prevalence and high blood pressure across U.S. states.

Model parameters

- y_{ij}: Age-adjusted prevalence of high blood pressure for state j in year i.
- x_{ij}: Scaled smoking prevalence for state j in year i, where a one-unit increase corresponds to a 5 percentage point increase in smoking prevalence.
- t_{ij}: Centered year variable for state j in year i, defined relative to the midpoint of the study period.
- α_j: State-specific intercept representing the typical high blood pressure prevalence for state j, after adjusting for smoking prevalence and time.
- μ_α: Global mean of the state-specific baselines across all states.
- σ_α: Standard deviation of the state-specific baselines, capturing how much typical high blood pressure levels vary between states.
- β_s: Global association between smoking prevalence and high blood pressure, where a one-unit increase corresponds to a 5 percentage point increase in smoking prevalence.
- β_t: Global linear time trend, representing the average change in high blood pressure prevalence over time.
- σ_y: Residual standard deviation capturing unexplained variation in high blood pressure prevalence at the state–year level.

- Each state has its own latent intercept, representing that state's typical high blood pressure level after accounting for smoking and time.
- These intercepts are drawn from a shared population distribution with two parameters: a mean baseline and a standard deviation that reflects how much states vary from each other.
- The model includes global slopes for:
  - Smoking prevalence
  - Year (linear time trend)
- The outcome variable (high blood pressure prevalence) is modeled as a normal distribution, centered on a linear combination of:
  - The state-specific baseline
  - The effect of smoking
  - The effect of time
- For each state and each year:
  - Global-level priors:
    - Global mean of state baselines: Normal(0, 5)
    - Standard deviation of state baselines: HalfNormal(5)
    - Smoking slope: Normal(0, 1)
    - Year slope: Normal(0, 1)
    - Residual standard deviation: HalfNormal(1)
  - State-level parameters: We use a non-centered parameterization to improve sampling:

        - Each state has a raw intercept drawn from Normal(0, 1)
        - The actual state baseline is computed as:
          baseline = global_mean + raw_intercept* global_sd
    - For each state-year:
        - The predicted blood pressure is:
          baseline + (smoking_slope * smoking_scaled) + (year_slope * year_centered)
        - The observed value is drawn from a normal distribution centered at this predicted value, with standard deviation equal to the residual SD.
    - Interpretation of Parameters:
        - State baseline: The typical high blood pressure level for a state, after adjusting for smoking and time.
        - Global mean baseline: The average across all states.
        - Standard deviation of baselines: How much states differ in their typical blood pressure levels.
        - Smoking slope: Change in blood pressure (in percentage points) associated with a 5 percentage point increase in smoking.
        - Year slope: Average change in blood pressure over time.
        - Residual SD: Remaining variation not explained by the model.
    - We use weakly informative priors that regularize the model without overwhelming the data. The prior for the global baseline mean is Normal(0, 5). Although centered at 0, the large amount of data pulls the posterior into the plausible 20–40% range for state-level prevalence.
        - The prior on the standard deviation of baselines (HalfNormal(5)) allows for many percentage points of variation across states, while still discouraging unrealistically large differences.
        - For the smoking and year slopes we use Normal(0, 1) priors. These do not assume a strong effect in either direction, but allow for moderate positive or negative associations if supported by the data.
        - The residual SD prior (HalfNormal(1)) keeps the noise level on a realistic scale while not remaining very informative.
- **Results**
  We estimate a clear positive association between state-level smoking prevalence and high blood pressure prevalence using a hierarchical Bayesian model with state-specific intercepts and a linear time trend. In the main specification, a 5 percentage point increase in smoking prevalence is associated with a posterior median increase of 1.17 percentage points in high blood pressure prevalence, with a 95% credible interval of [0.51, 1.84], indicating meaningful uncertainty but a consistently positive effect. To assess sensitivity to prior choice, we widened the prior on the smoking slope; the estimated association remained similar, with a posterior median of 1.34 and a 95% credible interval of [0.61, 2.02], suggesting the results are reasonably

robust to this hyperparameter choice. These findings should be interpreted as associative rather than causal, as the analysis relies on observational state-year data and does not account for other factors such as diet, obesity, income, or healthcare access.

- **Discussion**
  - Limitations and Future Directions:
    - Missing covariates: The model does not include other important predictors (such as obesity, diabetes, income) that could confound the smoking-blood pressure relationship
    - Assumes a linear relationship: The effect of smoking is assumed to be the same across all levels, which might not hold in reality
    - Same slope across states: We assume smoking affects all states equally. But in reality, the effect might differ depending on context (such as healthcare access or demographics)
  - Other Formulations/Graphical Models: We considered using a model with separate slopes for each state (random slopes), but the data were too limited to support that complexity. Those models produced unstable and not well-identified estimates, so we used a simpler and more reliable structure.
  - Potential Improvements/Additional Data:
    - Adding more predictors, such as socioeconomic status or healthcare access
    - Use finer-grained data (like county-level data) to build deeper hierarchical models
    - Try non-linear models to explore more complex relationships
  - Gao et al.'s study found a positive association between tobacco use and hypertension, which is consistent with our findings of a positive association between smoking prevalence and high blood pressure. Gao et al. analyze smoking at the individual level, whereas our study uses state-level smoking prevalence. This difference in measurement likely explains why our estimated association is smaller. Gao et al. also note that strength of smoking-hypertension can vary across populations, which aligns with our model's findings captured across states. Differences between Gao et al.'s model and ours can arise from aggregation effects, since state-level data can smooth over individual-level risks and short-term physiological responses to smoking. Our model does not incorporate the individual-level factors that Gao et al. discusses, which may account for some unexplained variation in high blood pressure across states. Overall, the association and biological plausibility found in our analysis is consistent with the patterns described by Gao et al.

7. CONCLUSIONS

Research Question 1:

- Outcomes Summary: Through our six pairwise comparisons of CVD mortality rates across our ozone exposure quartiles, we found that moderate ozone exposure was associated with higher CVD mortality than the lowest exposure group. Q4 is not as consistent with the quartiles but this is likely due to limited sample size and low statistical power, among other factors.
- Critical Evaluation
  1. Our analysis is limited by missing covariates such as weather conditions, socioeconomic factors, healthcare access, and co-pollutants. Additionally, using state-year aggregates rather than individual-level or daily data likely smooths short-term ozone spikes that may drive cardiovascular risk.
  2. It would be helpful to understand how ozone levels interact with other pollutants and chronic disease risk factors. A question we could ask a domain expert is: "To what extent do state-year ozone averages illustrate the effect of ozone levels on CVD and how can co-pollutants change this relationship?". The answer to this question can affect how we set up our problem, variables, and the model. For example, if the expert tells us that short-term spikes in ozone levels are more likely to drive CVD, we would need to redesign our metric for exposure. If multiple pollutants play a role in interacting with ozone, this would change our quartiles.
  3. Our conclusion is partially robust. While it shows that Q2 and Q3 being greater than Q1 is a consistent trend after both correction methods, Q4's lack of significance is an indication of low power. If we were to change the exposure metric from quartiles to a continuous model, it could show us non-linear relationships that were otherwise not present initially. A user should interpret the results as a pattern-based association, not as a definitive exposure-response curve. The quartile shows a general trend between ozone levels and CVD but it should not be over-interpreted.
  4. Our results are specific to the US and also represent state-year averages within the set time range. Because of this, the specifics should not be generalized to individuals or other countries. However, the general idea that lower ozone exposure can correlate with lower CVD mortality is a relationship that can potentially be generalized to other countries and other time periods.
- Recommendations
  1. A follow-up study using daily ozone exposure data linked to individual-level CVD outcomes would allow for stronger inference about short-term exposure effects and improve statistical power. Incorporating co-pollutants and weather variables would further clarify mechanisms.

2. Based on our results, we recommend that state and local public health agencies expand cardiovascular risk screening and public health advisories in regions with moderate ozone levels, rather than focusing solely on the highest-exposure areas. Specifically, agencies should prioritize these regions for enhanced CVD surveillance, targeted health messaging during elevated ozone periods, and coordination with air quality agencies to identify cost-effective emission reduction strategies. Acting at moderate exposure levels aligns with preventive public health goals and may reduce cardiovascular risk before ozone concentrations reach extreme thresholds.
3. Potential Impacts of this Call to Action:
   This recommendation is feasible because state and local public health departments and air quality agencies already have the authority and infrastructure to conduct CVD surveillance and issue targeted health advisories. However, residents, local businesses, or industry groups may reasonably object due to concerns about economic impacts or stigma associated with being labeled a higher-risk area. The benefits of this policy would be greatest for individuals in moderate ozone regions—especially older adults and those with existing cardiovascular risk—through earlier detection and prevention, while impacts on others would primarily be indirect. In response to objections, we emphasize that the recommendation focuses on supportive health services and education rather than regulation or penalties. Our call to action is guided by values of preventive public health and equity: acting early to reduce harm and ensuring that communities are supported rather than blamed for environmental risks. These values reflect our training in data-driven decision-making and a commitment to using evidence to inform policies that prioritize population health while minimizing unintended social or economic harm.

Research Question 2:

- Outcomes Summary: Through our Bayesian hierarchical model, we found a positive association between smoking and high blood pressure prevalence at the state-level across the US. A 5 percentage point increase in state-level smoking prevalence was associated with around 1.2 percentage points higher high blood pressure prevalence. The results are quite similar under a wider prior for β_smoke, which indicates that the general direction and magnitude of the association is due to the data rather than the prior choice made.
- Critical Evaluation
  1. There are several limitations in the data such as missing predictors (ex. socioeconomic factors, healthcare access, diet, etc.), the fact that our analysis is at the state-level rather than individual, and possible measurement errors due to the data potentially coming from surveys or self-reports.

2. In regards to domain knowledge, it would be helpful to know how quickly smoking shows up in population-level blood pressure or how much blood pressure variation can be attributed to smoking compared to other high blood pressure risk factors. A question that could be asked to a domain expert would be: "From your experience, how strong is the relationship between smoking patterns and high blood pressure at the state level, after accounting for other risk factors such as healthcare access, socioeconomic factors, and diet?" Knowing this answer could help us choose different or more covariates for our model. It would also inform us of how to phrase our conclusions in order to be the most accurate and not assume any direct correlations.
3. Overall, our conclusion appears to be quite robust as changing the prior and hyperparameter values does not appear to change our results/overall trends. However, our model does use a global linear slope for smoking across all states. This decision assumes that the relationship between smoking and blood pressure is linear and that it is homogenous across all states. We could potentially change this to a non-linear model that could show stronger effects when there are higher smoking levels and weaker effects when at lower smoking levels. This change has its own risks too (ex. overfitting). Given this potential bias within our model, users should treat our results as providing a general association across all states, rather than being specific to singular states/individuals. Higher smoking levels tend to be associated with increased high blood pressure prevalence, but the effect for a state specifically can vary.
4. Our results are specific to the US states and territories from 2011-2021 given the fact that we used CDI data. They also represent state-level data rather than individual-level. Given this information, they might not generalize to other countries, different time periods, and people on an individual level. However, the trends that we observe here may be generalizable to other US chronic diseases that have similar settings with similar risk factors.

- Recommendations
  1. A follow-up study that could build on our work is an individual-level, casual study that looks at how smoking directly affects high blood pressure risk among adults in the US. We can adjust for more confounding variables in this causal study. This study could confirm if our state-level results are also present at the individual level.
  2. Based on our results, we suggest that state-level hypertension care programs integrate smoking cessation programs, especially for programs that reside in states with high smoking prevalence. This is because our results indicate a positive association between high smoking prevalence and high blood pressure prevalence.
  3. Potential Impacts of this Call to Action:

a. This recommendation should be feasible as it won't be too hard to implement. Many states have established tobacco control programs as well as hypertension programs. The main action would be to integrate aspects of these programs' functions together. State health departments would have the power to redesign these programs in order to include the integrated functions. Clinic networks and other actors in the health system can also implement relevant tobacco screenings for those with chronic conditions like high blood pressure. Although this integration is meant to be a means of mitigating the concerns revolving around tobacco usage and high blood pressure, big tobacco companies may oppose these policies as they can affect tobacco sales.
b. Our recommendation is meant to benefit patients with high blood pressure who smoke/are exposed to smoking. Especially in communities where there is a high prevalence of both smoking and cardiovascular diseases, our recommendation will aim to provide these individuals with direct support. If an opponent claims that these programs might shame people who struggle with smoking and chronic disease (a reasonable claim), we can explain that our goal is to encourage health departments to divest more resources into smoking cessation programs as an integrated aspect of chronic disease care. There is no shame in taking part in these programs as in the long term, they will aim to reduce the suffering that these individuals are experiencing when it comes to both smoking and high blood pressure.
c. As a group, we recognize that smoking and chronic disease is often more prevalent in communities that are already experiencing other systemic inequities. In these communities, education about the effects of smoking may not be as widespread or simply ignored due to lack of resources and health education. Providing an integrated smoking cessation and chronic disease program should help to reduce these inequities and also spread awareness. From our experience (ex. Health classes at school), having adequate knowledge on health concerns and their effects plays an important role in the decisions we make. Knowing that an action (ex. smoking) can have a potential correlation to a chronic condition (ex. High blood pressure) can lead to better health choices being made.

Reference Page

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